IIA WEB DESIGN • UNIT 04

Organising Web Content

HTML links • lists • images • navigation • semantic structure

Bachelor of Library and Information Studies
Bachelor of Records Management Systems

HOME

SERVICES

CONTACT

What you will be able to do

By the end of Unit 4 you will be able to organise, connect and explain simple webpages.

Connect content

Create links to local pages, external sites and named sections.

Organise information

Use lists, images, captions and semantic page regions meaningfully.

Build a small site

Create a three-page library or records-centre example with working navigation.

KEY IDEA **Your website can work locally using HTML only; advanced JavaScript is not required.**

Why connected pages matter

A visitor should be able to move from an overview to the information they need.

Library visitor

Starts at Home, opens Services and checks borrowing guidance and opening hours.

Records-centre visitor

Starts at Home, reads public access procedures and locates contact details.

Design consequence

Consistent navigation and descriptive links reduce searching and confusion.

KEY IDEA A website is more than its homepage: the connections between pages matter.

Hyperlinks connect resources

The anchor element is the foundation of navigation on the Web.

Anchor element

<a> creates a clickable or keyboard-activatable link.

href attribute

Identifies the destination, such as a file path or a web address.

Visible link text

Tells visitors what they can expect when they follow the link.

HTML EXAMPLE

```
<a href="services.html">
  Explore library services
</a>
```

KEY IDEA The destination file must exist; HTML does not create it automatically.

Anatomy of an HTML link

Every working basic link has visible content and a meaningful destination.

Opening tag

 starts the hyperlink and supplies its destination.

Link text

The words inside the anchor communicate the purpose of the link.

Closing tag

 ends the hyperlink.

HTML EXAMPLE

```
<a href="pages/contact.html">
  Contact the records office
</a>
```

KEY IDEA Good text is descriptive: “Contact the records office” is clearer than “Click here”.

Four useful destinations

Links can point to websites, other files, page sections or documents.

External website

A full URL such as https://example.org may open a different website.

Local page

services.html points to a file in the same folder.

Section or document

#hours targets a named section; guide.pdf refers to a file.

HTML EXAMPLE

```
<a href="https://example.org">Help site</a>
<a href="services.html">Services</a>
<a href="#hours">Opening hours</a>
<a href="guide.pdf">Guide (PDF)</a>
```

KEY IDEA Choose labels that describe the destination and file type where helpful.

Write descriptive link text

Visitors should understand the purpose of a link without guessing from its surroundings.

Unclear

“Click here” or “Read more” tells very little when read alone.

Clear

“Download borrowing guidelines (PDF)” tells the user exactly what to expect.

Accessibility

Descriptive labels help keyboard users and people navigating a list of links.

HTML EXAMPLE

```
<!-- Less clear -->
<a href="hours.html">Click here</a>

<!-- Better -->
<a href="hours.html">
  Library opening hours
</a>
```

KEY IDEA Links should describe the resource or task rather than the act of clicking.

Absolute URL versus relative path

The same resource can be addressed through a complete URL or a local relationship.

Absolute URL

Includes the scheme and host: https://example.org/services.html.

Relative path

Names a file relative to the current HTML document: pages/services.html.

Choice

Use a complete URL for another site and relative paths for your own pages.

HTML EXAMPLE

```
<a href="https://example.org/help">
  External help
</a>
<a href="pages/services.html">
  Our services
</a>
```

KEY IDEA A relative path starts from the folder containing the current webpage.

Understand a website folder

The location of each file determines the relative path used to reach it.

Root file

index.html is in the main website folder.

Pages subfolder

services.html and contact.html are both inside pages/.

Assets folder

A local illustration is stored inside assets/img/.

SAMPLE FOLDER STRUCTURE

```
information-site/
├─ index.html
├─ pages/
│   ├─ services.html
│   └─ contact.html
└─ assets/
    └─ img/
        └─ reading-room.svg
```

KEY IDEA Keep an organised folder structure so paths remain predictable.

Paths from the homepage

A file in the main folder can link into its subfolders directly.

From index.html

To open a page in pages/, include that folder name in href.

For local images

To load a resource in assets/img/, include both subfolders in src.

Check case

The browser expects the same filenames, capitalisation and extension.

HTML EXAMPLE

```
<a href="pages/services.html">
  Services
</a>
<img src="assets/img/reading-room.svg"
     alt="Library reading room">
```

KEY IDEA **From the homepage use pages/services.html, not just services.html.**

Paths back to the homepage

Move up from pages/ to the website’s main folder with ../.

Current file

Imagine you are editing pages/services.html.

Destination

The homepage index.html is one folder above services.html.

Correct syntax

../index.html first moves up a folder, then finds index.html.

HTML EXAMPLE

```
<!-- Inside pages/services.html -->
<a href="../index.html">Home</a>
<a href="contact.html">Contact</a>
```

KEY IDEA Two files in the same folder can link to each other by filename only.

Avoid common path mistakes

A broken image or link usually means the browser cannot find its destination.

Missing folder

href="services.html" is incorrect on the homepage if the file is inside pages/.

Wrong number of ../

Moving up too many directories points outside the intended website.

Filename mismatch

Services.html and services.html may be treated differently on a server.

HTML EXAMPLE

```
<!-- Broken from index.html -->
<a href="services.html">Services</a>

<!-- Correct -->
<a href="pages/services.html">Services</a>
```

KEY IDEA Test links after each file move or rename.

Jump to a section with #id

A fragment link takes readers directly to a labelled part of a page.

Name the destination

Add a unique id to the heading or other element.

Link to that name

Use href="#hours" when the destination has id="hours".

Cross-page links

Use services.html#membership to reach a section on another page.

HTML EXAMPLE

```
<a href="#hours">Opening hours</a>

<h2 id="hours">Opening hours</h2>
<p>Monday–Friday, 08:00–17:00.</p>
```

KEY IDEA **The fragment name after # must match the destination id exactly.**

Email and telephone links

Special URL schemes can offer contact actions on supported devices.

Email contact

mailto: can open an installed mail application with a recipient address.

Telephone contact

tel: can offer a dialling action on supported phones.

Limit

These links do not process forms or send messages automatically.

HTML EXAMPLE

```
<a href="mailto:library@example.org">
  Email the library
</a>
<a href="tel:+255000000000">
  Call the office
</a>
```

KEY IDEA **Use demonstration addresses in practice websites, not personal contacts.**

Choose the correct kind of list

Lists communicate how information is related, not just how it is displayed.

Unordered list (ul)

Services with no required order: reading room, printing, assistance.

Ordered list (ol)

Steps with a meaningful sequence: read guidance, enquire, receive reply.

Description list (dl)

Terms paired with meanings: catalogue, archive, accession.

KEY IDEA Choose lists by meaning: bullets for sets, numbers for sequences, terms for definitions.

Unordered lists: ul and li

Use bullets when the sequence of items does not change their meaning.

Container

 marks a group of unordered list items.

Items

Each <li> contains one complete list item.

Professional example

A library homepage can list services available to all visitors.

HTML EXAMPLE

```
<h2>Library services</h2>
<ul>
  <li>Reading spaces</li>
  <li>Reference support</li>
  <li>Internet access</li>
</ul>
```

KEY IDEA Avoid using line breaks alone to imitate a list; use semantic list elements.

Ordered lists: ol and li

Use numbers where the order of the steps is significant.

Container

 tells the browser that items have a meaningful sequence.

Items

 marks the first step, second step and subsequent steps.

Professional example

Public records enquiries often involve a clearly explained procedure.

HTML EXAMPLE

```
<h2>Public enquiry process</h2>
<ol>
  <li>Check access guidance.</li>
  <li>Identify the type of record.</li>
  <li>Contact the office.</li>
</ol>
```

KEY IDEA An ordered list is better for procedures; an unordered list is better for categories.

Description lists: dl, dt and dd

A description list connects terms with their definitions or explanations.

dl

Groups related terms and descriptions.

dt

Identifies each term, such as Catalogue or Archive.

dd

Provides a related definition or explanation.

HTML EXAMPLE

```
<dl>
  <dt>Catalogue</dt>
  <dd>Describes library resources.</dd>
  <dt>Archive</dt>
  <dd>Records kept for continuing value.</dd>
</dl>
```

KEY IDEA **Description lists work well for small glossaries and service terms.**

Insert an image with img

An image is a separate resource whose file path is supplied through src.

Image source

src specifies the image file to display.

Alternative text

alt communicates relevant information for people who cannot see the image.

No closing tag

img is a void element; write <img ...>, not <img>...</img>.

HTML EXAMPLE

```
<img src="assets/img/reading-room.svg"
     alt="Library reading room with shelves"
     width="500" height="250">
```

KEY IDEA A local image must be included in the project folder if the website is to work offline.

Write useful alternative text

Alternative text should reflect the informational role of the image.

Informative image

“Entrance to archives reading room beside reception” explains what the image shows.

Decorative image

Use alt="" if an ornament provides no meaningful information.

Poor example

“image123.jpg” gives a filename, not the image’s meaning.

HTML EXAMPLE

```
<img src="entry.jpg"
     alt="Entrance to the archives room">

<img src="border.svg" alt="">
```

KEY IDEA **Alt text is for meaning, not a list of visual details unrelated to the page.**

Image dimensions and responsive layouts

Image size attributes help preserve layout, while CSS will later support adaptability.

Width and height

Reserve a layout area while an image loads, reducing sudden movement.

Do not stretch blindly

A fixed size does not guarantee good display on every screen.

Loading

loading="lazy" can delay off-screen images until needed.

HTML EXAMPLE

```
<img src="reading-room.jpg"
     alt="Study tables in the library"
     width="640" height="360"
     loading="lazy">
```

KEY IDEA For responsive image styling later, CSS can use max-width: 100% and height: auto.

Figures and visible captions

A caption and alternative text serve different users and purposes.

figure

Groups an illustration, photograph or diagram with its caption.

figcaption

Adds visible contextual information under the figure.

alt

Provides an image alternative; a caption does not automatically replace it.

HTML EXAMPLE

```
<figure>
  <img src="room.svg"
       alt="Library reading tables">
  <figcaption>
    Example reading-room facilities.
  </figcaption>
</figure>
```

KEY IDEA **Describe what matters for understanding the image in its page context.**

Respect licences and confidentiality

Public information websites must consider image ownership and record sensitivity.

Image permission

Create your own visuals or use assets with suitable permission or licensing.

Privacy

Never publish identifiable borrowers or protected records in a student project.

Fictional data

Practice sites should use invented institutions and service details.

KEY IDEA A public service page should explain access procedures, not expose confidential records.

Semantic HTML explains page roles

Meaningful elements describe the structure of a webpage.

header and footer

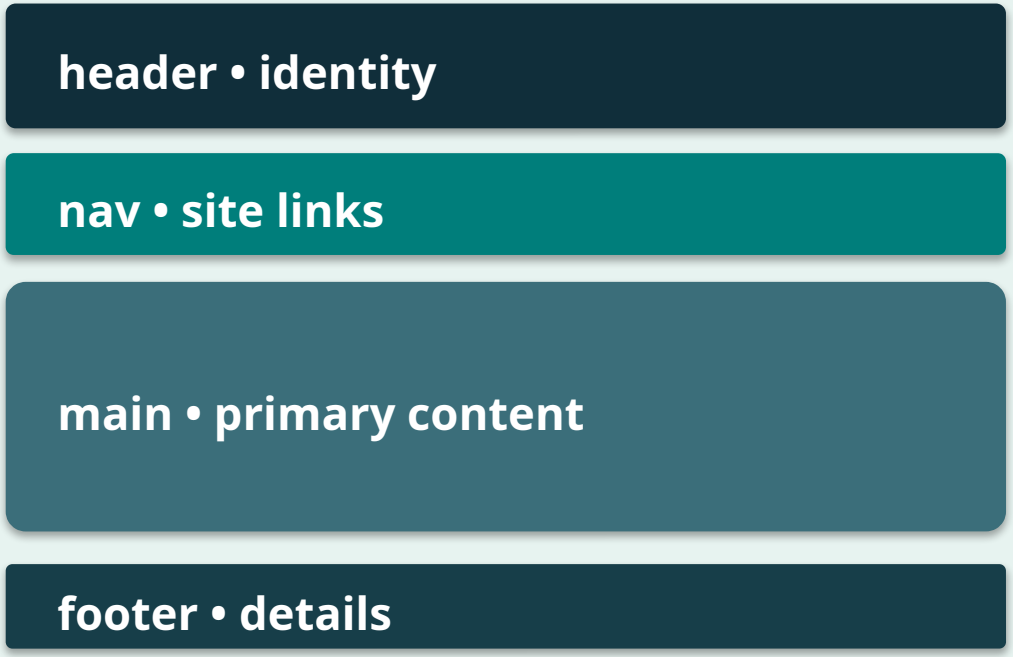
Introduce a page or section and provide supporting details.

nav

Groups the links used for navigation.

main

Contains the primary content unique to the current webpage.



KEY IDEA **Semantic regions help visitors, assistive technology and future maintainers.**

A simple semantic HTML layout

Use structural elements around headings, paragraphs and navigation links.

Header

Contains the site title and introductory information.

Main

Holds the content that distinguishes the page from other pages.

Footer

Provides supporting contact information or a short notice.

HTML EXAMPLE

```
<header><h1>Community Library</h1></header>
<nav aria-label="Main navigation">
  <a href="index.html">Home</a>
</nav>
<main><h2>Welcome</h2></main>
<footer><p>Public information.</p></footer>
```

KEY IDEA One document normally has one primary <main> landmark.

Sections, articles and headings

A meaningful heading helps readers understand what follows.

section

Groups related content under a topic, such as Opening hours.

article

Represents self-contained information, such as an announcement.

Heading hierarchy

Use h1 for the page title and h2 for major subsections.

HTML EXAMPLE

```
<main>
  <section>
    <h2>Opening hours</h2>
    <p>Monday–Friday, 08:00–17:00.</p>
  </section>
</main>
```

KEY IDEA **Use semantic regions for meaning, not simply to change appearance.**

Plan a small three-page website

The site map shows how a homepage links to two useful destinations.

Home

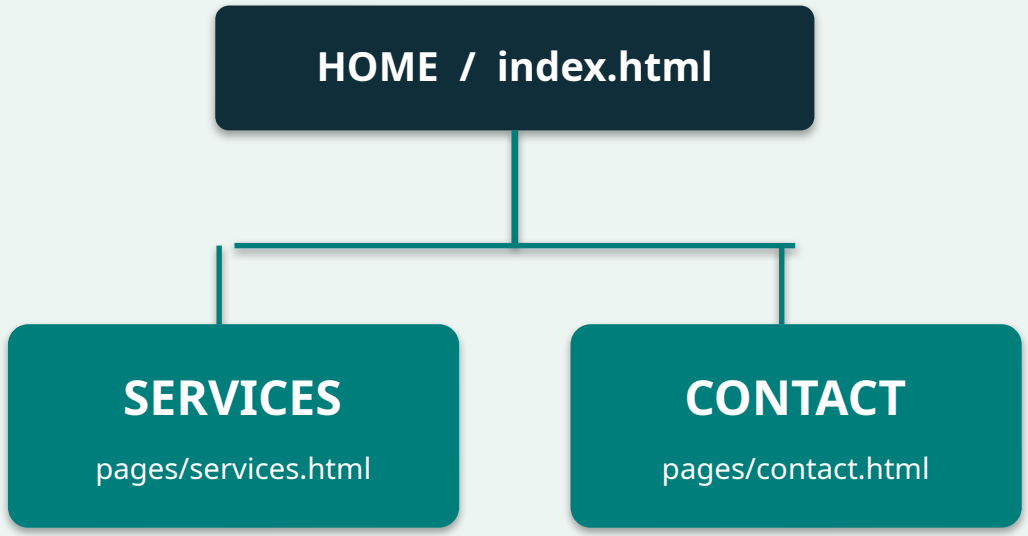
Introduces the institution and links to important services.

Services

Explains what users can do or which public procedures to follow.

Contact

Gives reliable service hours and public contact information.



KEY IDEA A simple predictable structure is more useful than many empty pages.

Consistent navigation across pages

Use similar menus so visitors can move easily between resources.

Same labels

Home, Services and Contact should mean the same things on every page.

Correct file paths

From pages/services.html, ../index.html returns to Home.

Current page

aria-current="page" identifies which navigation item is active.

HTML EXAMPLE

```
<nav aria-label="Main navigation">
  <a href="../index.html">Home</a>
  <a href="services.html"
     aria-current="page">Services</a>
  <a href="contact.html">Contact</a>
</nav>
```

KEY IDEA Test every menu link after adding or renaming files.

Library application: service navigation

A library website helps readers move from a general welcome to specific assistance.

Homepage

Welcomes visitors and links to services, hours and contact details.

Services page

Lists reading support and uses ordered steps to explain borrowing.

Image and caption

A reading-room illustration has informative alt text and a visible caption.

HTML EXAMPLE

```
<h2>Library services</h2>
<ul>
  <li>Reading areas</li>
  <li>Reference support</li>
</ul>
<a href="pages/services.html">
  See all library services
</a>
```

KEY IDEA **Visitors should locate opening hours and borrowing guidance without searching blindly.**

Records-centre application: public access

Records pages can explain public procedures without exposing protected information.

Homepage

Introduces available public enquiry services and how to find guidance.

Services page

Uses an ordered list to explain a general request procedure.

Privacy

Contains fictional examples, not actual personnel files or restricted references.

HTML EXAMPLE

```
<h2>Enquiry steps</h2>
<ol>
  <li>Read the access guidance.</li>
  <li>Identify the record category.</li>
  <li>Contact the records office.</li>
</ol>
```

KEY IDEA Clear procedures support access; confidentiality rules still apply.

Guided practical: test three linked pages

Use the downloadable starter website to see navigation and file paths in action.

Open

Extract the Unit 4 starter ZIP; open your chosen index.html locally.

Edit

Change the fictional institution title and one item in each services list.

Check

Use Home, Services and Contact links; verify the image and page headings.

KEY IDEA **The starter includes separate library and records-centre websites, both working offline.**

Assignment 4: keep it simple

Develop a small two-page site that demonstrates links and lists.

Homepage

Create index.html with one institution heading and a short introduction.

Services page

Create services.html with three services in an unordered list.

Navigation and evidence

Link the pages both ways; prepare a ZIP of two HTML files and one screenshot.

KEY IDEA CSS, JavaScript, databases and real personal information are not required.

Check your understanding

Explain these three ideas before completing the shuffled quiz.

Links

What do the href attribute and visible link text each do?

Paths

Why is ../index.html often needed when linking back from pages/services.html?

Images and semantics

When should an image have useful alt text, and what does <main> mean?

KEY IDEA **The online quiz gives a score and explains incorrect or unanswered answers.**

Unit 4: essential takeaways

You can now organise content into a small, connected information-service site.

Content connections

Anchor links join webpages and named sections; relative paths follow folders.

Useful organisation

Lists, images, captions and meaningful page regions improve comprehension.

Professional application

A well-navigated library or records-centre site serves real information needs.

KEY IDEA Next: Unit 5 — HTML Tables for catalogue and records-inventory information.